
KronHiPPO-STGP: Online Bayesian Spatio-temporal Modelling with Evolving Memory

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Abstract

Online Bayesian inference requires updating posterior beliefs while storing only a bounded summary of historical observations. Existing streaming approaches compress past data into finite states, but these states are usually constructed under a fixed representation. When latent parameters are updated or inducing coordinates change, the stored state may no longer correspond to the original likelihood contribution. We study this issue in online spatio-temporal Gaussian processes with evolving finite-dimensional representations. We propose KronHiPPO-STGP, a method that maintains a bounded Gaussian likelihood state for a parametric mean component and a residual Gaussian process. The proposed state supports posterior updates after representation changes by transferring likelihood factors between successive HiPPO inducing coordinates without replaying previous observations. We derive three properties of evolving Bayesian memory: the cross information required to update coupled trend and residual components, the finite-dimensional projection operator between changing representations together with its associated information loss, and the conditions under which likelihood transfer preserves Kronecker structure. Based on these results, we develop a Schur–Sylvester posterior recovery procedure that is equivalent to solving the corresponding projected Gaussian system while avoiding dense matrix operations. Experiments on environmental, public-health, and traffic datasets show that KronHiPPO-STGP maintains memory cost independent of stream length and achieves accurate prediction with calibrated uncertainty under online representation changes.

1 INTRODUCTION

Online Bayesian models must repeatedly update predictions as new observations arrive while discarding raw historical data. This requirement creates a fun-

damental tension: inference should preserve sufficient historical evidence to maintain uncertainty, while the memory state must remain bounded as the stream grows.

Gaussian processes (GPs) provide a natural framework for uncertainty-aware spatio-temporal modelling because they maintain coherent distributions over latent functions and capture dependencies across space and time (Rasmussen and Williams, 2006; Hartikainen et al., 2011; Hamelijnck et al., 2021). However, exact GP inference requires computation and storage that grow with the number of observations. Practical online deployment therefore relies on finite representations, such as inducing variables or approximate likelihood summaries (Titsias, 2009; Matthews et al., 2016; Csató and Oppé, 2002; Bui et al., 2017; Maddox et al., 2021).

Sparse variational GPs and streaming GP methods address this challenge by replacing raw history with finite posterior or likelihood summaries. These approaches answer an important question: *How can historical observations be compressed?* However, online inference faces an additional difficulty when the finite representation used to store historical evidence evolves over time: *How can compressed Bayesian evidence remain valid after the representation changes?* A finite Bayesian summary is defined with respect to a particular inducing representation. When this representation changes, the stored statistics may no longer correspond to the same latent quantities.

A common strategy in spatio-temporal Gaussian process modelling is to decompose observations into a parametric trend component and a correlated residual process:

$$y = \phi^\top \beta + f.$$

The trend component captures systematic variations explained by covariates, while the residual GP models remaining structured spatio-temporal uncertainty (Hamelijnck et al., 2021). This decomposition is particularly relevant in online Bayesian inference because historical residual evidence depends on the parametric component used when the data were compressed.

Therefore, revising the mean parameters changes the interpretation of previously stored residual information.

To maintain long-range temporal information with bounded memory, recent work has explored structured temporal representations such as HiPPO. HiPPO provides a principled mechanism for compressing the history of a signal into a fixed-dimensional state and has become a foundation for long-range sequence models such as S4 and Mamba (Gu et al., 2020, 2022; Gu and Dao, 2023).

Building on this idea, Recurrent Memory for Online Interdomain Gaussian Processes introduced HiPPO as a temporal interdomain inducing representation for Gaussian processes, demonstrating that HiPPO-based inducing variables can retain long-range temporal information with a fixed number of variables (Chen et al., 2025). Inspired by these results, we adopt HiPPO inducing representations for spatio-temporal Gaussian processes, enabling compact storage of long-range temporal information without increasing the number of temporal inducing variables.

However, unlike conventional inducing variables whose interpretation remains fixed, HiPPO inducing representations evolve as the temporal horizon changes. The projection functions used to summarize historical information are updated as new observations arrive (Gu et al., 2020; Chen et al., 2025). Consequently, two inducing states may have identical dimensions while representing different GP functionals. Thus, fixed dimension does not imply fixed semantics. This raises a fundamental Bayesian inference question: *How can historical likelihood information remain valid when the inducing representation itself changes?*

This evolving representation creates three challenges for bounded Bayesian memory. First, posterior revision of latent parameters requires preserving the coupling between trend and residual components. A residual-only memory cannot correctly update historical evidence after mean revision. Second, changing HiPPO representations require likelihood transport rather than direct state reuse. Historical Bayesian statistics must be projected into the new inducing representation. Third, the transferred Bayesian state must preserve exploitable structure; otherwise recovering the posterior over product spatio-temporal inducing variables requires dense computation.

We introduce KronHiPPO-STGP, a bounded-state Bayesian inference framework for online Gaussian processes with evolving finite representations. The maintained state is a joint Gaussian likelihood factor over the parametric mean and residual inducing variables. It preserves three types of information: (1) revision information that allows historical residual evidence to be

corrected after mean updates; (2) transportable likelihood statistics that can be projected between successive HiPPO inducing representations; and (3) structured residual factors that enable scalable posterior recovery.

Our contributions are:

- **Joint online inference with HiPPO memory.** We construct a fixed-size Gaussian natural-parameter state that jointly updates the linear mean coefficients and spatio-temporal GP residual, and identify the trend-residual cross-precision block required to revise historical residual contributions without replay.
- **Characterised transfer and structured posterior solves.** We quantify the one-step loss incurred when the historical approximate-likelihood factor is projected into the updated HiPPO representation. Under the stated assumptions, the transferred likelihood retains a Kronecker structure, allowing the projected Gaussian posterior to be recovered through Schur-Sylvester equations without any additional approximation beyond the finite-dimensional inducing-representation projection.
- **Predictive and mechanism evidence.** Across three real-world domains, KronHiPPO-STGP improves the main predictive metrics over the evaluated online GP baselines. Controlled interventions test the two main components of the retained natural-parameter state: prediction degrades substantially when joint trend-residual information is removed or when historical evidence is carried forward without accounting for changing HiPPO inducing representations. Conditional transport remains close to a reference posterior recomputed in the current inducing representation, while matched solver experiments demonstrate the computational benefit of the preserved Kronecker structure.

2 RELATED WORK

Finite GP representations and online memory. Sparse variational methods define finite representations through inducing variables while retaining the GP conditional (Titsias, 2009; Matthews et al., 2016), and interdomain methods extend these variables to linear functionals (Lázaro-Gredilla and Figueiras-Vidal, 2009; van der Wilk et al., 2020). Streaming GPs instead compress past observations into posterior or approximate-likelihood factors (Csató and Oppner, 2002; Bui et al., 2017; Maddox et al., 2021), with memory-based variants retaining selected observations to mitigate sequential approximation (Chang et al., 2023). These approaches

provide finite-dimensional posterior or approximate-likelihood summaries and permit changes in the inducing representation, but they do not characterise the cross information needed when the posterior over an uncertain parametric trend is revised. OHSVGP introduces time-dependent HiPPO interdomain inducing variables for long-term online memory (Gu et al., 2020; Chen et al., 2025), and subsequent work derives an analytic HiPPO-LegS interdomain covariance construction for the same temporal-memory setting (Guo et al., 2026). This prior work establishes how temporal GP memories can be constructed and evolved, but not how joint trend-residual likelihood information should be revised and transferred in a spatio-temporal model, or when the resulting residual likelihood retains sufficient structure for scalable posterior computation.

Structure for spatio-temporal GPs. State-space methods exploit Markov structure along time (Hartikainen and Särkkä, 2010; Nickisch et al., 2018), while doubly sparse and ST-SVGP models combine temporal state-space structure with variational spatial sparsity (Adam et al., 2020; Hamelijnck et al., 2021). Kronecker methods exploit separability on Cartesian supports (Van Loan, 2000; Flaxman et al., 2015). Latent-Kronecker inference can accommodate missing observations in a fixed batch Cartesian-product grid (Lin et al., 2025). In our setting, by contrast, the interdomain approximate-likelihood factor itself changes over time, so the transferred residual precision must retain the factorisation required for structured online posterior solves.

Positioning. We build on streaming likelihood transfer between inducing representations (Bui et al., 2017) and time-dependent HiPPO interdomain inducing variables (Chen et al., 2025). We consider joint online inference for an uncertain parametric mean and a separable spatio-temporal GP whose temporal inducing functionals change over time. The method retains the trend-residual cross-precision information needed for replay-free revision, quantifies the loss induced by finite-dimensional conditional-expectation projection of the approximate log-likelihood factor, and identifies conditions under which the projected residual likelihood preserves the single-Kronecker structure needed for scalable posterior computation.

3 PRELIMINARIES

We introduce only the objects needed to state the replay-free inference problem and the finite target transported by the method.

Online observations. At block k , the observations at visible locations V_k are available before predictions are made at held-out locations Q_k . The held-out values are revealed only after prediction, and processed raw blocks are not replayed. This defines contemporaneous spatial nowcasting; dataset-specific calibration and evaluation details are given in Section 5 and Appendix B.

Trend-residual model and finite VFE target. We model the observation as

$$y(t, s) = \phi(t, s)^\top \beta + f(t, s) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_y^2). \quad (1)$$

The coefficient β is a static latent variable, $\beta \sim \mathcal{N}(m_{\beta,0}, S_{\beta,0})$. Online learning updates its posterior but does not posit coefficient drift. The residual GP has separable kernel

$$k((t, s), (t', s')) = k_t(t, t')k_s(s, s'). \quad (2)$$

Changing HiPPO interdomain coordinates. HiPPO memory stores weighted integrals of the past residual process. As the integration horizon expands, the weighting functions change while the number of stored coefficients remains fixed. Let $\mathcal{H}_k = [t_0, t_k]$ be the expanding temporal horizon. For spatial inducing locations z_j , the mixed interdomain variables are

$$u_{m,j}^{(k)} = \int_{\mathcal{H}_k} f(t, z_j) \psi_m(t; \mathcal{H}_k) dt, \quad m = 0, \dots, M_t - 1, \quad j = 1, \dots, M_s. \quad (3)$$

The vector $u_k = \text{vec}(U_k)$ therefore has fixed dimension, but u_{k-1} and u_k are different GP functionals because their HiPPO bases are normalised on different horizons. Its prior and observation map are

$$p(u_k) = \mathcal{N}(0, K_{t,k} \otimes K_s), \quad A_k = P_{t,k} \otimes C. \quad (4)$$

Here A_k maps the inducing state to the observations. These variables define only the finite variational representation; the underlying GP conditional is retained rather than replaced by a deterministic inducing approximation. We therefore use

$$p(f_k | u_k) = \mathcal{N}(A_k u_k, K_{f_k f_k} - Q_{f_k f_k}), \quad (5)$$

$$q_k(\beta, u_k, f_k) = p(f_k | u_k) q_k(\beta, u_k). \quad (6)$$

For fixed (β, u_k) , taking expectation over the retained GP conditional $p(f_k | u_k)$ gives

$$\begin{aligned} \mathbb{E} \|y_k - \Phi_k \beta - f_k\|^2 \\ = \|y_k - \Phi_k \beta - A_k u_k\|^2 + \text{tr}(K_{f_k f_k} - Q_{f_k f_k}). \end{aligned} \quad (7)$$

The first term supplies the Gaussian natural statistics, while the second is the standard VFE trace correction (Titsias, 2009). Prediction retains the GP conditional residual $\nu_* = k_{**} - k_{*u} K_u^{-1} k_{u*}$, so uncertainty is not restricted to the finite inducing span.

4 METHOD

Each online update first projects the retained likelihood statistics into the current HiPPO representation, then assimilates the newly visible observations and recovers the joint posterior over the mean coefficients and residual process. Figure 1 summarises the construction. The following sections explain which statistics must be retained, characterise the projection error, and establish the structure used by the solver.

4.1 Revision-aware memory

For a fixed latent coordinate $z = [\beta^\top, u^\top]^\top$ and design $J_j = [\Phi_j, A_j]$, the Gaussian variational factor at fixed hyperparameters has

$$\Lambda_k = \Lambda_0 + \sigma_y^{-2} \sum_{j \leq k} J_j^\top J_j, \quad h_k = h_0 + \sigma_y^{-2} \sum_{j \leq k} J_j^\top y_j. \quad (8)$$

Thus fixed-coordinate sequential accumulation is identical to optimising the same finite VFE factor over concatenated blocks. Its off-diagonal precision contains the trend-residual cross block $R_{\beta u}$, equivalently $R_{u\beta} = R_{\beta u}^\top$, with $R_{u\beta} = \sigma_y^{-2} \sum_j A_j^\top \Phi_j$.

The role of this block can be seen without invoking a broad convergence claim. If old residual evidence was formed using $\hat{\beta}$, define

$$h_u^{<k}(\hat{\beta}) = \sigma_y^{-2} \sum_{j \leq k} A_j^\top (y_j - \Phi_j \hat{\beta}).$$

Then a revised estimate $\hat{\beta}' = \hat{\beta} + \Delta_\beta$ gives

$$h_u^{<k}(\hat{\beta}') - h_u^{<k}(\hat{\beta}) = -R_{u\beta}^{<k} \Delta_\beta. \quad (9)$$

A residual-only summary is stale whenever the right-hand side is nonzero. Once the historical blocks have been discarded, the correction can be reconstructed only if the retained state encodes the action of $R_{u\beta}$. The cross block therefore records the historical information required to reinterpret the compressed residual statistic after the posterior mean estimate is revised.

Proposition 1 (Necessity of trend-residual cross information). *Fix the historical designs and a reference coefficient β_0 . Suppose a replay-free summary, together with $h_u^{<k}(\beta_0)$, must recover $h_u^{<k}(\beta_0 + \Delta)$ without error for every admissible $\Delta \in \mathbb{R}^{d_\beta}$. Then the summary must determine the linear map $\Delta \mapsto R_{u\beta}^{<k} \Delta$; when the admissible changes span $\mathbb{R}^{d_\beta}$, this determines $R_{u\beta}^{<k}$ itself. Hence retaining $R_{u\beta}$, or an information-equivalent encoding of its action, is necessary for faithful correction after arbitrary posterior trend revision.*

Proposition 1 formalises this information requirement: faithful replay-free revision must retain $R_{u\beta}$, or an

information-equivalent representation of its action. The proof is given in Appendix A.1.

4.2 Coordinate-consistent likelihood transfer

Successive HiPPO variables u_o and u_n have equal dimension but represent different GP functionals, and replacing one representation by the other need not be an invertible reparameterisation. We therefore project the historical log-likelihood factor into the current representation while keeping it separate from the prior expressed in the current coordinates. Write the retained historical factor as

$$\tilde{\ell}_o(\beta, u_o) \propto \exp \left\{ -\frac{1}{2} z_o^\top R_o z_o + r_o^\top z_o \right\}. \quad (10)$$

Let $L_{on} = K_{on} K_{nn}^{-1}$ denote the conditional-mean map from the new GP functionals to the old ones, and let $M_{on} = \text{diag}(I, L_{on})$. Taking the conditional expectation of $\log \tilde{\ell}_o(\beta, u_o)$ under $p(u_o | u_n)$ gives a quadratic log factor in (β, u_n) .

Proposition 2 (Joint historical-likelihood transport). *Define*

$$\log \tilde{\ell}_{o \rightarrow n}(\beta, u_n) = \mathbb{E}_{p(u_o | u_n)} [\log \tilde{\ell}_o(\beta, u_o)] + \text{const}, \quad (11)$$

where $K_{nn} \succ 0$. Then the non-constant natural parameters are

$$R_{o \rightarrow n} = M_{on}^\top R_o M_{on}, \quad r_{o \rightarrow n} = M_{on}^\top r_o. \quad (12)$$

In particular, $R_{\beta u} \leftarrow R_{\beta u} L_{on}$ and $R_{uu} \leftarrow L_{on}^\top R_{uu} L_{on}$.

This transformation preserves the quadratic natural-parameter form required by the online recursion. Appendix A.2 gives the derivation. The projection is generally lossy because the new finite state need not retain all information represented by the old GP functionals; Section 4.3 quantifies this loss.

4.3 What finite transfer cannot preserve

Conditional expectation gives the L^2 -optimal representation of the historical log factor among functions measurable in (β, u_n) , but the projection is not generally lossless. Components of the old log likelihood that are not measurable in (β, u_n) cannot be represented after transfer. Proposition 3 quantifies this one-step loss.

Assume $K_{nn} \succ 0$, $R_{uu} = R_{uu}^\top$, and finite Gaussian second moments. Let $\Sigma_{o|n} = K_{oo} - K_{on} K_{nn}^{-1} K_{no}$ and partition the old quadratic factor as in Eq. (10).

Proposition 3 (One-step projection optimality and loss). *For $g(\beta, u_o) = \log \tilde{\ell}_o(\beta, u_o)$ up to an additive*

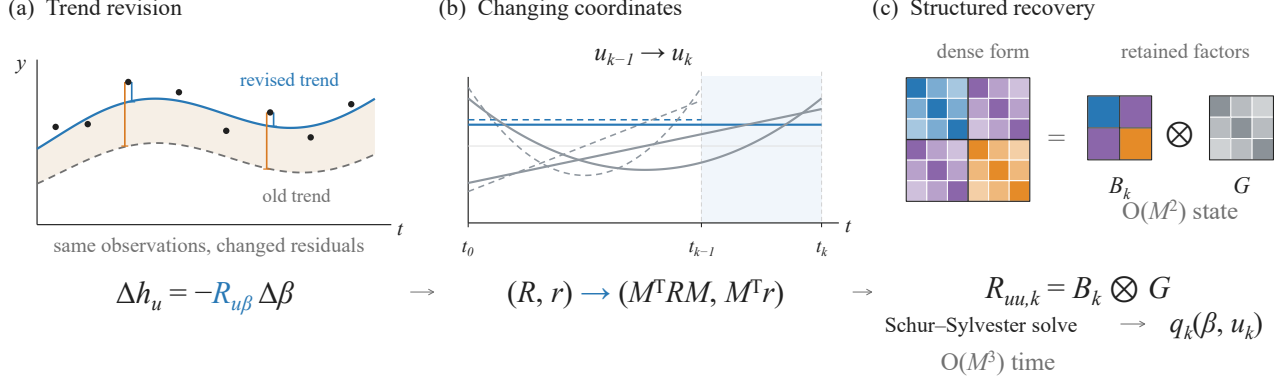


Figure 1: Three stages of bounded-state inference. (a) Posterior trend revision changes the residual contribution of historical observations, making the retained trend-residual cross statistic necessary for replay-free correction. (b) Successive HiPPO states represent different GP functionals, so conditional projection transports historical likelihood evidence between changing coordinates. (c) The single-Kronecker likelihood invariant enables structured recovery of the transferred finite target without dense product-state factorisation. The complexity annotation assumes $M_s \asymp M_t = M$ and fixed d_β under Proposition 4.

constant, $\mathbb{E}[g \mid \beta, u_n]$ is its optimal L^2 approximation measurable in (β, u_n) . With

$$a = r_u - R_{u\beta}\beta - R_{uu}L_{on}u_n,$$

the conditional projection error is

$$\begin{aligned} \mathbb{E}[(g - \mathbb{E}[g \mid \beta, u_n])^2 \mid \beta, u_n] \\ = a^T \Sigma_{o|n} a + \frac{1}{2} \text{tr}[(R_{uu} \Sigma_{o|n})^2]. \end{aligned} \quad (13)$$

The result is deliberately local: it characterises the best one-step representation in the new finite state and does not imply monotone or globally lossless repeated transfer. The proof, operator-norm bound, and nested-space corollary are given in Appendix A.3.

4.4 Structure-preserving transport

For bounded-state online inference, the transferred likelihood must remain both statistically meaningful and computationally structured. Under a separable residual kernel and fixed spatial geometry, the coordinate change acts only on the temporal factor. With fixed spatial observations and inducing functionals, C is constant and $G = C^T C$, while $L_{k-1,k} = L_t \otimes I_{M_s}$ with $L_t = K_{k-1,k}^{(t)} (K_{k,k}^{(t)})^{-1}$.

Proposition 4 (Single-Kronecker likelihood invariant). *Assume a separable residual kernel, unchanged spatial inducing and observation functionals, Cartesian blocks, and homoscedastic Gaussian noise. If $R_{uu,k-1} = B_{k-1} \otimes G$, then conditional transport followed by assimilation gives*

$$\begin{aligned} B_k &= L_t^T B_{k-1} L_t + \sigma_y^{-2} P_{t,k}^T P_{t,k}, \\ R_{uu,k} &= B_k \otimes G, \end{aligned} \quad (14)$$

Starting from $B_0 = 0$, the invariant holds at every block.

The proof is in Appendix A.4. Adding the prior produces the two-term residual posterior operator

$$D_u = K_{t,k}^{-1} \otimes K_s^{-1} + B_k \otimes G. \quad (15)$$

The joint precision is

$$\Lambda_k = \begin{bmatrix} A_\beta & B_{\beta u} \\ B_{\beta u}^T & D_u \end{bmatrix}. \quad (16)$$

The persistent state is $\{R_{\beta\beta,k}, R_{\beta u,k}, B_k, r_{\beta,k}, H_{u,k}, K_{t,k}\}$, together with fixed model quantities, where $r_{u,k} = \text{vec}(H_{u,k})$. It does not materialise the $(M_s M_t)^2$ residual block. Thus the stream length disappears from the state dimension: only the fixed trend block, cross statistics, and temporal/spatial factors persist.

4.5 Structured recovery of the transferred target

Propositions 1–4 determine what historical information must be retained, how it is projected between changing coordinates, what finite projection can lose, and which separable structure survives. It remains to recover the resulting coupled Gaussian posterior without materialising the $M_s M_t$ -dimensional residual precision. The standard Schur complement reduces joint recovery to applications of D_u^{-1} and one d_β -dimensional solve. We therefore focus on the nontrivial residual solve, whose Kronecker structure admits the following matrix-equation form.

Lemma 1 (Structured Sylvester solve of the residual block). *For symmetric positive-definite $K_s, K_{t,k}$ and*

positive-semidefinite G, B_k , the system $D_u \text{vec}(Z) = \text{vec}(Q)$ is equivalent to

$$K_s^{-1} Z K_{t,k}^{-1} + G Z B_k = Q. \quad (17)$$

After whitening and separately diagonalising the spatial and temporal factors, the matrix equation decouples elementwise,

$$\hat{Z}_{ij} = \frac{\hat{Q}_{ij}}{1 + \gamma_i \eta_j}. \quad (18)$$

The transformations and recovery map are given in Appendix A.5.

Preserving the residual likelihood factorisation makes this classical matrix-equation solve applicable at every online update. The derivation and predictive covariance calculations are given in Appendix A.5.

Theorem 1 (Equivalence to dense recovery of the transferred target). *Under the assumptions of Proposition 4, proper Gaussian priors, fixed hyperparameters, and declared numerical regularisation, the Schur-complement recovery together with Lemma 1 produces the same posterior means, covariance quadratic forms, and marginal predictive variances as densely solving the same precision Λ_k in Eq. (16), up to floating-point error and jitter.*

The proof is in Appendix A.6. Theorem 1 separates representation error from numerical recovery: historical likelihood projection defines the finite approximation, and Schur–Sylvester recovery solves that projected Gaussian system exactly up to the declared numerical regularisation. For fixed mean dimension and balanced spatial and temporal budgets, recovery uses $O(M^3)$ time and $O(M^2)$ persistent memory; the general expressions are given in Appendix A.8.

Algorithm 1 KronHiPPO-STGP: replay-free update

Require: stream $\mathcal{D}_{1:K}$ with (V_k, Q_k) ; budgets (M_s, M_t) ; prior $p(\beta)$.
1 $\theta \leftarrow \text{CALIBRATE}(\mathcal{D}_1)$; $\text{FREEZE}(\theta)$
2 $\mathcal{S} \leftarrow \text{INITIALISEBOUNDEDSTATE}()$
3 **for** $k = 2, \dots, K$ **do**
4 Observe $y_k(V_k)$
5 $L_t \leftarrow \text{HiPPOTRANSFER}(\mathcal{H}_{k-1}, \mathcal{H}_k; \theta)$
6 $\mathcal{S} \leftarrow \text{ASSIMILATE}(\text{TRANSPORT}(\mathcal{S}, L_t), \mathcal{D}_k[V_k])$
7 $q_k(\beta, u_k) \leftarrow \text{SCHURSILVESTER}(\mathcal{S})$
8 $(\mu_k, v_k) \leftarrow \text{PREDICT}(q_k, Q_k)$ // before reading $y_k(Q_k)$
9 **end for**
Return: predictions and bounded state $\mathcal{S}_K$.

Algorithm 1 exposes the execution order and strict-online information boundary; Propositions 2–4 and the Schur–Sylvester solve define its core operations. Here $\mathcal{S}_k$ is the bounded persistent likelihood state defined above, containing $R_{\beta\beta,k}$, $R_{\beta u,k}$, B_k , $r_{\beta,k}$, $H_{u,k}$, and $K_{t,k}$.

5 EXPERIMENTS

We organise the experiments around four questions motivated by the analysis: (i) does the full construction transfer across domains? (ii) does retaining joint trend–residual information matter in practice? (iii) must historical evidence be updated when the HiPPO representation changes, and how accurate is the resulting finite-state transfer? and (iv) does the preserved Kronecker structure translate into measurable computational savings?

Unless stated otherwise, all online experiments use the same information boundary: visible locations are observed, held-out locations are predicted before their labels are revealed, and previously processed blocks are never replayed. The main comparison reports RMSE, CRPS, Gaussian NLPD, and ECE; protocols, coverage, per-split results, and additional diagnostics are provided in the appendix. Table 1 anchors the cross-domain predictive comparison. Figure 2 shows representative PEMS-BAY trajectories, while Figure 3 mirrors the three stages in Figure 1 with controlled checks of revision-aware coupling, coordinate-consistent transfer, and structured posterior recovery. Complete per-split, calibration, transfer-capacity, and solver statistics are reported in Appendix B.

Baselines. We compare against streaming sparse GP methods, structured spatio-temporal and multi-output GP models, and the task-aligned IGNNK neural kriging model on PEMS-BAY. OHSVGP provides a comparison with existing HiPPO-based GP memory, while Kron-STGP serves as our fixed-coordinate control. Method references are given in Table 1, and Appendix B.2 records the archived implementation provenance.

5.1 Cross-domain replay-free prediction

We first test the same bounded-state inference construction across three qualitatively different processes. ERA5-Land provides the longest environmental stream and uses a comparatively rich meteorological mean representation; COVID-19 introduces weekly public-health regime changes; and PEMS-BAY tests urban mobility with a substantially simpler mean design. PEMS-BAY is evaluated as contemporaneous held-out-sensor prediction rather than standard future traffic forecasting, making IGNNK a task-aligned external kriging baseline (Li et al., 2018; Wu et al., 2021). Together, the three settings test portability of the inference construction rather than performance of one domain-specific forecasting architecture.

Table 1 shows consistent performance of the same replay-free construction across all three domains.

Table 1: Cross-domain strict-online spatial prediction (mean $\pm$ SD; five splits for ERA5-Land/COVID-19 and three for PEMS-BAY). Bold and underlined values denote the best and second-best comparable results; ‘–’ indicates unavailable metrics. PEMS-BAY uses contemporaneous held-out-sensor prediction.

Method	RMSE $\downarrow$	CRPS $\downarrow$	NLPD $\downarrow$	ECE $\downarrow$
<i>ERA5-Land</i>				
OSGPR (Bui et al., 2017)	0.9833 $\pm$ 0.0127	0.5612 $\pm$ 0.0086	1.5847 $\pm$ 0.0294	0.1131 $\pm$ 0.0051
StreamingSGPR (Maddox et al., 2021)	1.0660 $\pm$ 0.0196	0.6283 $\pm$ 0.0144	1.7331 $\pm$ 0.0466	0.1543 $\pm$ 0.0080
OHSVGP (Chen et al., 2025)	0.7303 $\pm$ 0.0302	0.4132 $\pm$ 0.0170	1.2989 $\pm$ 0.0686	<u>0.1063$\pm$0.0091</u>
Kron-STGP	<u>0.4166$\pm$0.0046</u>	<u>0.2391$\pm$0.0016</u>	<u>0.6590$\pm$0.0051</u>	0.1711 $\pm$ 0.0018
KronHiPPO-STGP	0.1416$\pm$0.0036	0.0766$\pm$0.0017	−0.3349$\pm$0.0434	0.0669$\pm$0.0067
<i>COVID-19</i>				
Persistence	0.1955 $\pm$ 0.0095	0.1087 $\pm$ 0.0051	−0.1894 $\pm$ 0.0321	<u>0.0656$\pm$0.0316</u>
Adaptive OSGPR (Bui et al., 2017)	0.1742 $\pm$ 0.0290	0.0990 $\pm$ 0.0167	−0.2603 $\pm$ 0.1757	<u>0.1267$\pm$0.0435</u>
ST-SVGP (Hamelijnck et al., 2021)	0.2094 $\pm$ 0.0136	0.1192 $\pm$ 0.0067	−0.0840 $\pm$ 0.0390	0.0900 $\pm$ 0.0239
LMC-SVGP (Jeong and Kim, 2023)	0.3376 $\pm$ 0.0529	0.1965 $\pm$ 0.0281	1.0809 $\pm$ 0.4905	0.1806 $\pm$ 0.0140
ICM-SVGP (Jeong and Kim, 2023)	0.3435 $\pm$ 0.0447	0.2002 $\pm$ 0.0219	1.0005 $\pm$ 0.3588	0.1797 $\pm$ 0.0016
FSDE-SVI (Jeong and Kim, 2023)	0.4705 $\pm$ 0.0478	0.2580 $\pm$ 0.0158	0.6693 $\pm$ 0.1044	0.0238$\pm$0.0066
Kron-STGP	<u>0.1605$\pm$0.0107</u>	<u>0.0922$\pm$0.0055</u>	−0.3029 $\pm$ 0.0533	0.1427 $\pm$ 0.0240
KronHiPPO-STGP	0.1580$\pm$0.0120	0.0885$\pm$0.0058	−0.3621$\pm$0.0583	<u>0.1183$\pm$0.0343</u>
<i>PEMS-BAY</i>				
IGNNK (Wu et al., 2021)	0.7968 $\pm$ 0.1226	–	–	–
OHSVGP	0.8190 $\pm$ 0.0681	<u>0.4064$\pm$0.0204</u>	<u>1.2312$\pm$0.0760</u>	<u>0.1671$\pm$0.0124</u>
OSGPR	0.8207 $\pm$ 0.0685	0.4130 $\pm$ 0.0211	1.2318 $\pm$ 0.0652	0.1816 $\pm$ 0.0083
Kron-STGP	0.8139 $\pm$ 0.0695	0.4104 $\pm$ 0.0220	1.2242 $\pm$ 0.0664	0.1827 $\pm$ 0.0096
KronHiPPO-STGP	0.7666$\pm$0.0629	0.3774$\pm$0.0193	1.1883$\pm$0.1076	0.1317$\pm$0.0196

KronHiPPO-STGP attains the lowest mean RMSE and the lowest comparable CRPS and NLPD, although the calibration metrics do not produce a uniform ranking. The cross-domain pattern is therefore more informative than dominance on every individual metric. On PEMS-BAY, the improvement also persists outside the high-dimensional meteorological mean design used on ERA5 and against the task-aligned IGNNK baseline.

Figure 2 provides a qualitative view across the three predeclared PEMS-BAY difficulty regimes. In these displayed trajectories, IGNNK smooths several abrupt congestion transitions that KronHiPPO-STGP tracks more closely while also reporting predictive uncertainty. We treat these panels as illustrations rather than aggregate evidence; the quantitative comparison is reported in Table 1.

5.2 Mechanism validation: memory and transfer

Revision-aware memory. Figure 3(a) tests the practical role of the trend–residual cross-precision block identified in Proposition 1. Within each setting, every joint/changing and zero-cross/changing pair uses the same spatial split, frozen hyperparameters, inducing-variable budgets, and changing-coordinate updates. Zero-cross removes the accumulated trend–residual cross block after the transported historical and current-

block statistics have been combined; all other retained statistics and assimilation steps are unchanged. Removing this block increases RMSE at both ERA5 operating points and on PEMS-BAY. At the archived Table-1 ERA5 operating point, RMSE rises from 0.1416 to 1.2921, while the full arm reproduces the archived predictions to numerical precision. This operating-point replication shows that the effect is not specific to the spectral-mixture mechanism configuration. Because the intervention removes both historical and current-block cross-block contributions, it tests joint mean–residual likelihood coupling rather than historical trend revision in isolation. Complete predictive and calibration metrics are reported in Appendix Table 6; calibration does not yield a uniform ordering.

Coordinate identity versus transfer. Figure 3(b) tests whether equal-dimensional HiPPO states can be treated as the same GP representation. Identity reuse carries the historical approximate-likelihood natural parameters forward unchanged while updating the HiPPO inducing representation; the prior, current-block update, hyperparameters, and posterior solver are held fixed. At $M_t = 128$, conditional transport gives predictive KL 1.26×10^{-4} to the reference posterior recomputed in the current representation, whereas identity reuse gives 2.07. The mismatch propagates to prediction: identity reuse raises RMSE from 0.0795 to 0.2651 and reduces Coverage90 from 0.982 to 0.632. CRPS

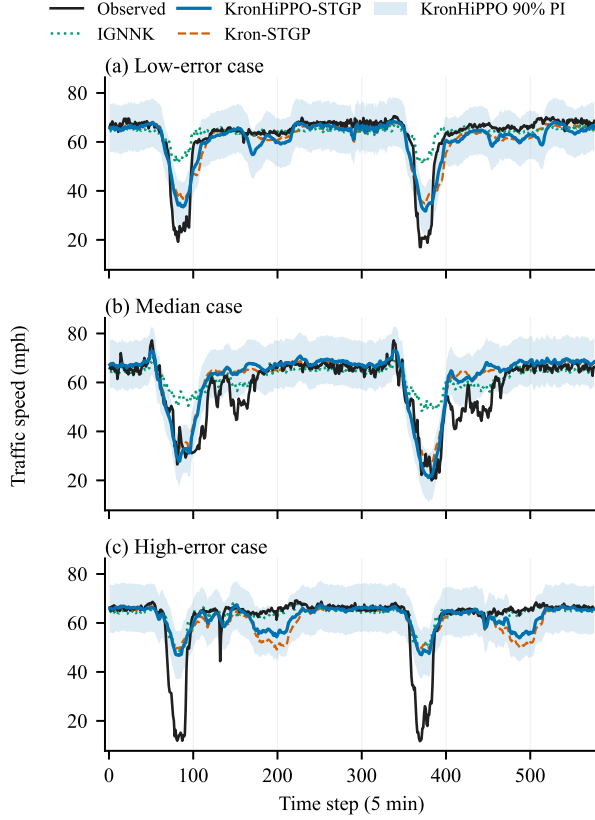


Figure 2: PEMS-BAY predictions across three predeclared difficulty regimes. The panels show seed-1 sensors nearest the 15th/50th/85th percentiles of KronHiPPO-STGP RMSE. Lines compare KronHiPPO-STGP with IGNNK and Kron-STGP; shading shows the 90% predictive interval. Sensor IDs and selection details are reported in Appendix B.

and NLPD worsen in the same comparison, while ECE moves in the opposite direction, so we do not claim a uniform calibration ordering. Complete statistics are reported in Appendix Table 7.

Transfer fidelity. The capacity sweep separates finite-projection error from coordinate mismatch. Conditional transport is non-monotone in M_t , but predictive KL falls to 3.67×10^{-3} at $M_t = 64$ and 1.26×10^{-4} at $M_t = 128$, whereas identity reuse remains near order one for $M_t \geq 32$. Increasing temporal capacity alone therefore does not correct coordinate mismatch; at the operating capacity, conditional transport closely matches the current-representation finite-dimensional target. We do not infer a general convergence guarantee from this sweep. Appendix Fig. 7 reports the complete capacity diagnostic.

5.3 Structured posterior computation and efficiency

Figure 3(c) tests the computational consequence of the preserved Kronecker structure. Dense and Schur–Sylvester recovery solve the same positive-definite systems with matched factors, right-hand sides, and BLAS threading. The mean paired speed-up rises from $7.9\times$ at $M_s = M_t = 16$ to $1036.7\times$ at $M_s = M_t = 64$, while relative solution error remains below 5×10^{-12} . For balanced budgets ($M_s = M_t = M$) and fixed d_β , the widening gap is consistent with the $O(M^3)$ structured and $O(M^6)$ dense solve complexities. Because both methods target the same Gaussian system, the speed-up is not obtained by changing the posterior approximation. Selected timings and relative errors are reported in Appendix Table 8.

The matched solver benchmark is the evidence for the algorithmic claim. A separate archived RTX 4090 implementation audit includes full-model overhead: at $M_s = M_t = 128$, KronHiPPO-STGP records $110.8\times$ lower training wall time and $8.8\times$ lower peak reserved memory than ST-SVGP, with 5.4% higher RMSE. These are implementation-level measurements rather than architecture-neutral complexity ratios; the $40.4\times$ profiler-count difference is likewise implementation-specific. Appendix B.7 reports the full audit and long-stream accounting.

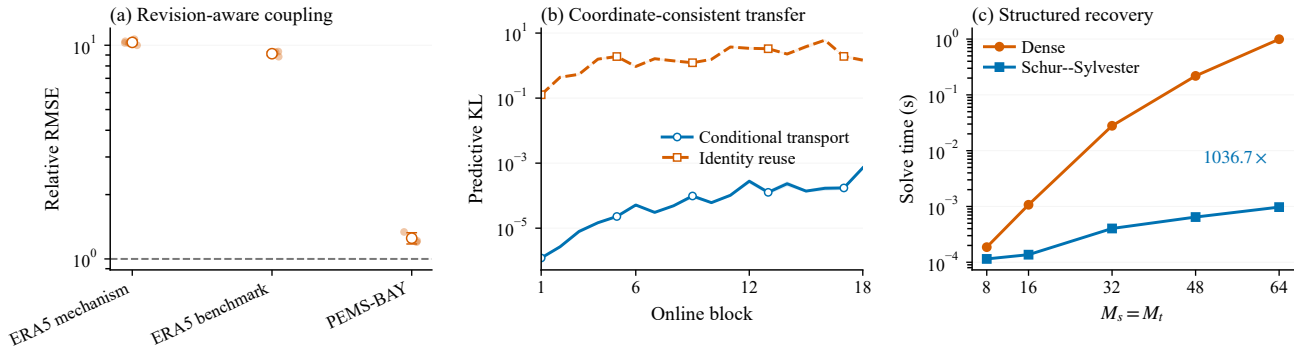


Figure 3: Controlled checks aligned with the three stages of Figure 1. (a) Removing the trend-residual cross-precision block increases RMSE under two ERA5-Land operating points and on PEMS-BAY; points show the per-split ratio ($\text{RMSE}_{\text{zero-cross}}/\text{RMSE}_{\text{joint}}$) under changing HiPPO coordinates, with (1) denoting no change. (b) Conditional transport keeps the online predictive distribution close to the current-representation variational reference, whereas identity reuse produces a persistent mismatch. (c) Schur-Sylvester recovery solves the same Gaussian system increasingly faster than dense recovery as the balanced product-state dimension grows. Error bars in (a) and bands in (b) denote sample standard deviations over the corresponding matched splits. Detailed predictive and calibration metrics and selected solver timings are reported in Appendix B.6.

6 CONCLUSION

We introduced KronHiPPO-STGP, a bounded-memory variational GP method for jointly updating a parametric mean and spatio-temporal GP residual without replay. Across environmental, public-health, and traffic streams, the method provides strong point and probabilistic predictions while keeping persistent memory independent of stream length. The analysis and controlled experiments examine each component of the construction: retaining the joint information needed to revise past evidence, updating that evidence when the HiPPO representation changes, and preserving Kronecker structure for efficient posterior computation.

The current construction assumes separable residual kernels, fixed spatial inducing and observation geometry, Cartesian-product observation blocks with Gaussian likelihood, and frozen streaming hyperparameters; richer observation patterns and likelihoods will require more general structured states. The results also indicate that, in online probabilistic models, a changing finite memory should be treated as a changing coordinate system for historical evidence rather than as an interchangeable latent state.

AI use statement

We used OpenAI ChatGPT and Codex as research-support and editing tools. Their use included suggesting literature-search queries and organising candidate sources; assisting with revisions to experimental code and additions to training scripts; organising experiment records and checking the consistency of tables

and reported results; improving LaTeX layout, cross-references, and figure and table formatting; and suggesting language and structural revisions to selected sections. Generative-AI outputs were treated as suggestions rather than scholarly evidence. The authors checked cited claims against the original publications, reviewed and tested code changes, generated the reported numerical results by running the documented experiments, and verified mathematical arguments and reported values against project records. All material was selected, revised, and approved by the authors, who remain responsible for its accuracy and originality.

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KronHiPPO-STGP: Online Bayesian Spatio-temporal Modelling with Evolving Memory: Supplementary Materials

REPRODUCIBILITY STATEMENT

The appendix specifies the information boundary, data splits, objectives, hyperparameter-freezing rule, metrics, and structured-solver check. The anonymised supplementary package contains machine-readable aggregate and per-seed mechanism results, the suite manifest, proof audit, and scripts used to run and visualise the diagnostics.

Ethics statement

The study uses environmental observations and aggregate public-health data from public sources. It does not recruit participants, conduct clinical interventions, or access identifiable or individual-level health records. The models were evaluated as research methods and were not deployed for clinical or public-health decision making. Any operational use would require separate assessment of calibration, distribution shift, fairness, and the consequences of missed or false alarms.

A DERIVATIONS AND EXACTNESS BOUNDARY

A.1 Proof of Proposition 1: necessity of the cross statistic

Let S denote the retained summary and assume a zero-error decoder $D(S, h_u(\beta_0), \Delta) = h_u(\beta_0 + \Delta)$ for every admissible Δ . Equation (9) then implies

$$h_u(\beta_0) - D(S, h_u(\beta_0), \Delta) = R_{u\beta}\Delta.$$

The retained information therefore determines the action of $R_{u\beta}$ on the span of admissible changes. If this span is $\mathbb{R}^{d_\beta}$, evaluating the decoder on a basis recovers every column of $R_{u\beta}$. Conversely, retaining $R_{u\beta}$ makes the correction explicit through Eq. (9). Two historical datasets with the same retained summary but different actions $R_{u\beta}\Delta$ would require the decoder to return different values from identical inputs, contradicting error-free recovery.

A.2 Proof of Proposition 2: historical likelihood-factor transfer

Conditionally write $u_o = L_{on}u_n + \delta$ with $\mathbb{E}[\delta \mid u_n] = 0$. For $z_o = M_{on}z_n + [0, \delta]^\top$, substitution into Eq. (10) followed by conditional expectation gives

$$-\frac{1}{2}z_n^\top M_{on}^\top R_o M_{on} z_n + r_o^\top M_{on} z_n - \frac{1}{2} \text{tr}(R_{uu,o} \text{Cov}(\delta)).$$

The trace is constant in (β, u_n) , leaving Eq. (12). The transferred cross block is therefore $R_{\beta u, o} L_{on}$. This establishes a one-step projected log factor and does not compare the resulting posterior with a full-history posterior expressed in the current inducing representation.

A.3 Proof of Proposition 3: one-step projection optimality and loss

Write $u_o = m + \delta$, where $m = L_{on}u_n$ and, conditional on (β, u_n) , $\delta \sim \mathcal{N}(0, \Sigma_{o|n})$. Expanding the terms that depend on δ gives

$$\begin{aligned} g(\beta, m + \delta) &= c(\beta, u_n) + a^\top \delta - \frac{1}{2} \delta^\top R_{uu} \delta, \\ a &= r_u - R_{u\beta} \beta - R_{uu} m. \end{aligned}$$

After centring,

$$g - \mathbb{E}[g \mid \beta, u_n] = a^\top \delta - \frac{1}{2} \{ \delta^\top R_{uu} \delta - \text{tr}(R_{uu} \Sigma_{o|n}) \}.$$

The cross covariance vanishes because centred Gaussian third moments are zero. The linear term has variance $a^\top \Sigma_{o|n} a$, while the unscaled quadratic term has variance $2 \text{tr}[(R_{uu} \Sigma_{o|n})^2]$. This gives Eq. (13), and optimality follows because conditional expectation is the orthogonal projection in L^2 . Finally, $a^\top \Sigma a \leq \|\Sigma\|_2 \|a\|_2^2$ and $\text{tr}[(R\Sigma)^2] = \|\Sigma^{1/2} R \Sigma^{1/2}\|_F^2 \leq \|R\|_F^2 \|\Sigma\|_2^2$.

For the nested-space statement, the conditional mean is again an L^2 projection. Its squared reconstruction error is the trace of the conditional covariance, and the Pythagorean theorem for nested projection subspaces makes this error non-increasing. This statement requires both the old target and its joint law to remain fixed; unrelated feature families at different M_t need not be nested.

A.4 Proof of Proposition 4: single-Kronecker likelihood invariant

Under separability and unchanged spatial functionals, $K_{k-1,k} = K_{k-1,k}^{(t)} \otimes K_s$ and $K_{k,k} = K_{k,k}^{(t)} \otimes K_s$. Hence

$$\begin{aligned} L_{k-1,k} &= K_{k-1,k} K_{k,k}^{-1} \\ &= \{K_{k-1,k}^{(t)} (K_{k,k}^{(t)})^{-1}\} \otimes I_{M_s} \\ &= L_t \otimes I_{M_s}. \end{aligned}$$

If $R_{uu,k-1} = B_{k-1} \otimes G$, conditional transport gives

$$(L_t \otimes I)^\top (B_{k-1} \otimes G) (L_t \otimes I) = (L_t^\top B_{k-1} L_t) \otimes G.$$

For a Cartesian block, $A_k = P_{t,k} \otimes C$ and therefore

$$\sigma_y^{-2} A_k^\top A_k = (\sigma_y^{-2} P_{t,k}^\top P_{t,k}) \otimes (C^\top C).$$

With $G = C^\top C$, their sum is precisely $B_k \otimes G$. The base case $B_0 = 0$, followed by induction over blocks, establishes Proposition 4.

A.5 Proof of Lemma 1: structured Sylvester solve for the inducing-variable precision block

Under the column-major vectorisation convention, $(A \otimes B) \text{vec}(Z) = \text{vec}(BZA^\top)$. Applying this identity to the two terms in Eq. (15) gives Eq. (17). Substitute

$$Z = K_s^{1/2} \tilde{Z} K_{t,k}^{1/2}$$

and multiply by the corresponding square roots. This gives

$$\tilde{Z} + \tilde{G} \tilde{Z} \tilde{B} = \tilde{Q},$$

where

$$\begin{aligned} \tilde{G} &= K_s^{1/2} G K_s^{1/2}, & \tilde{B} &= K_{t,k}^{1/2} B K_{t,k}^{1/2}, \\ \tilde{Q} &= K_s^{1/2} Q K_{t,k}^{1/2}. \end{aligned}$$

Let

$$\tilde{G} = U_s \text{diag}(\gamma_i) U_s^\top, \quad \tilde{B} = U_t \text{diag}(\eta_j) U_t^\top,$$

and define

$$\hat{Z} = U_s^\top \tilde{Z} U_t, \quad \hat{Q} = U_s^\top \tilde{Q} U_t.$$

Writing $\tilde{Z} = U_s \hat{Z} U_t^\top$ and rotating by these eigensystems gives $(1 + \gamma_i \eta_j) \hat{Z}_{ij} = \hat{Q}_{ij}$. Positive semidefiniteness implies $1 + \gamma_i \eta_j \geq 1$, establishing Eq. (18) and uniqueness. The recovered matrix is $\tilde{Z} = U_s \hat{Z} U_t^\top$ followed by $Z = K_s^{1/2} \tilde{Z} K_{t,k}^{1/2}$.

A.6 Proof of Theorem 1: dense equivalence of the transferred finite-dimensional target

Proposition 4 gives the lower-right block D_u of the densely assembled transferred precision. Lemma 1 replaces each application of D_u^{-1} with an algebraically equivalent structured matrix solve, after which the standard Schur-complement block inverse is applied to the same Λ_k . Define the Schur complement

$$P_\beta = A_\beta - B_{\beta u} D_u^{-1} B_{\beta u}^\top.$$

Then

$$\begin{aligned} S_{\beta\beta} &= P_\beta^{-1}, & S_{\beta u} &= -P_\beta^{-1} B_{\beta u} D_u^{-1}, \\ S_{uu} &= D_u^{-1} + D_u^{-1} B_{\beta u}^\top P_\beta^{-1} B_{\beta u} D_u^{-1}. \end{aligned}$$

These are precisely the blocks of Λ_k^{-1} , so the posterior means and covariance quadratic forms agree with the dense posterior solve. Adding the same full-GP conditional residual to both predictive variances preserves this equality. Any remaining difference arises only from finite-precision arithmetic and the same declared jitter; the structured solve introduces no additional posterior approximation.

A.7 Prediction with the full conditional

Let a_* denote the conditional-mean projection row, $h_* = [\phi_*^\top, a_*^\top]^\top$, and let S_k be the joint variational covariance. The predictive moments are

$$\mathbb{E}[y_*] = \phi_*^\top m_\beta + a_*^\top m_u, \quad (19)$$

$$\text{Var}(y_*) = \sigma_y^2 + h_*^\top S_k h_* + \nu_*, \quad \nu_* = k_{**} - a_*^\top K_u a_*. \quad (20)$$

The ν_* term is the GP conditional variance term and is distinct from the VFE trace correction used during hyperparameter calibration.

A.8 Detailed posterior-solve cost

A dense inducing-variable precision factorisation costs $O(M_s^3 M_t^3)$ time and $O(M_s^2 M_t^2)$ memory. Accounting for the d_β cross-block right-hand sides and the Schur complement, structured posterior solve costs

$$O(M_s^3 + M_t^3 + (d_\beta + 1)(M_s^2 M_t + M_s M_t^2) + d_\beta^2 M_s M_t + d_\beta^3), \quad (21)$$

and the persistent state occupies

$$O(M_s^2 + M_t^2 + (d_\beta + 1)M_s M_t + d_\beta^2), \quad (22)$$

for fixed budgets, independently of the number of elapsed blocks. These expressions cover core posterior computation and persistent state, and exclude current-block statistic formation, data loading, Task-1 calibration, and all-point prediction.

B EXPERIMENTAL PROTOCOLS

B.1 Evaluation metrics

For N held-out predictions with Gaussian posterior $y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, define $z_i = (y_i - \mu_i)/\sigma_i$. We report

$$\text{RMSE} = \left[\frac{1}{N} \sum_{i=1}^N (y_i - \mu_i)^2 \right]^{1/2}, \quad (23)$$

$$\text{NLPD} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1}{2} \log(2\pi\sigma_i^2) + \frac{(y_i - \mu_i)^2}{2\sigma_i^2} \right], \quad (24)$$

$$\text{CRPS}_i = \sigma_i \left[z_i \{2\Phi(z_i) - 1\} + 2\phi(z_i) - \frac{1}{\sqrt{\pi}} \right], \quad (25)$$

$$\text{CRPS} = \frac{1}{N} \sum_{i=1}^N \text{CRPS}_i, \quad (26)$$

where Φ and ϕ denote the standard-normal distribution and density. Calibration is measured over the central predictive intervals $\mathcal{A} = \{0.05, 0.15, \dots, 0.95\}$:

$$\hat{C}(\alpha) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left(|y_i - \mu_i| \leq \Phi^{-1} \left(\frac{1+\alpha}{2} \right) \sigma_i \right), \quad (27)$$

$$\text{ECE} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} |\hat{C}(\alpha) - \alpha|. \quad (28)$$

All four metrics are lower-is-better. Tables report the mean and sample standard deviation across the archived spatial splits. For archived sample-based predictive distributions, the same central-interval definition is evaluated using empirical predictive quantiles.

For the ERA5-Land and PEMS-BAY mechanism suites, complete mean $\pm$ sample SD results, including Coverage90, are reported in Appendix Table 6.

B.2 Dataset and protocol overview

Table 5 summarises the spatial units, temporal stream, information boundary, and role of each dataset. Dataset-specific figures are shown in the corresponding sections to keep each visual legible at print size.

Table 5: Dataset and strict-online protocol overview.

Dataset	Units	Stream	V / H	Domain
ERA5-Land	1,000 UK	171 blk.	800/200	Environmental
COVID	52 juris.	143 wk.	42/10	Public health
PEMS	325 sens.	5-min	260/65	Traffic

Baseline implementations. Table 1 compares streaming sparse GP methods, structured spatio-temporal and multi-output GP baselines, OHSVGP as an existing HiPPO-based GP method, and IGNNK as the task-aligned neural kriging baseline. The provenance table below includes only information recoverable from the archived code, manifests, and result records; unavailable fields are marked as not recorded. OSGPR and Adaptive OSGPR use the same official `OSGPR_VFE` core but differ in their archived update policies. StreamingSGPR denotes the same `online_gp` implementation used in Table 1 and Appendix Table 10.

Archived baseline implementation provenance: implementation and objective.

Method	Implementation	Objective
OSGPR	Official <code>OSGPR_VFE</code> (Bui et al., 2017)	Streamed VFE log factor
Adaptive OSGPR	Same official core; adaptive wrapper	VFE with causal parameter updates
StreamingSGPR	Official <code>online_gp</code> code path (Maddox et al., 2021)	Fantasy-model conditioning
OHSVGP	Official <code>HIPPOSVGP</code> multidimensional model (Chen et al., 2025)	Online variational HiPPO GP
ST-SVGP	Archived causal-refit adapter (Hamelijnc et al., 2021)	Gaussian ELBO
LMC-SVGP	FactorialSDE LMC constructor (Jeong and Kim, 2023); GPflow SVGP (Hensman et al., 2015)	Gaussian SVGP ELBO
ICM-SVGP	FactorialSDE intrinsic-coreionalisation constructor (archive id <code>imc_svgp</code>) (Jeong and Kim, 2023); GPflow SVGP (Hensman et al., 2015)	Gaussian SVGP ELBO
FSDE-SVI	FactorialSDE implementation (Jeong and Kim, 2023)	State-space variational ELBO
IGNNK	Official <code>basic_structure.IGNNK</code> (Wu et al., 2021)	Masked-node MSE

Archived baseline implementation provenance: budgets, inputs, and calibration.

Method	Inducing/state budget	Mean/covariates	Calibration
OSGPR	ERA5: 2×64 ; PEMS: 4×32	Task-1-fitted residual offset	Task-1 kernel/noise frozen
Adaptive OSGPR	COVID: $8 \times 32 = 256$	Archived Task-1 offset	25 Task-1 steps/block; 5 online steps
StreamingSGPR	ERA5: $2 \times 64 = 128$	Task-1-fitted residual offset	Task-1 kernel/noise frozen; 20% resampling
OHSVGP	ERA5: $M = 32, 128$ RFF; PEMS: $M = 64, 256$ RFF	Task-1-fitted residual offset	Task-1 kernel/noise mapped and frozen; variational state updated online
ST-SVGP	COVID: 32 spatial inducing points	Target series only	Task-1 fit; causal weekly refits
LMC-SVGP	not recorded for the reported three-run aggregate	Target series only	not recorded in returned aggregate
ICM-SVGP	not recorded for the reported three-run aggregate	Target series only	not recorded in returned aggregate
FSDE-SVI	not recorded for the reported three-run aggregate	Target series only	not recorded in returned aggregate
IGNNK	hidden width 100; diffusion order 1; 24-step window	Visible speeds, lagged held-out speeds, and fixed road graph	Task-1-only training; validation every 25 epochs

B.3 ERA5-Land

The processed dataset contains 2-m dew-point temperature and associated meteorological covariates from the ERA5-Land hourly product (Muñoz Sabater, 2019; Muñoz-Sabater et al., 2021) at 1,000 UK locations. Five paired spatial splits contain 800 visible and 200 held-out locations. Within each split, the validation locations used during Task-1 calibration are drawn from the visible set and remain disjoint from the 200 held-out evaluation locations. The mean design contains seven deterministic time-coordinate features and 126 current, lagged, and difference features from six non-target meteorological variables; the target itself is not used as an X-lag input. In the benchmark archive, OSGPR, StreamingSGPR, and OHSVGP receive deterministic Task-1-fitted mean offsets constructed from this feature design, whereas Kron-STGP and KronHiPPO-STGP retain the same features within their joint mean update. Both ERA5 configurations use the 186 Task-1 timestamps for VFE calibration. The scored stream begins at block 0 of Task 2 and contains 171 blocks across Tasks 2–10. The benchmark configuration uses the default temporal kernel, with at most 250 calibration iterations, validation every five iterations, and patience eight. The mechanism configuration uses a three-component spectral-mixture temporal kernel with 100 fixed calibration iterations. Both use $M_s = M_t = 128$ and 256 random Fourier features. The archive names `task1_10` and `task1_2` refer to experiment configurations rather than the temporal extent of the calibration input. The joint posterior state, including $q_k(\beta)$, is updated blockwise.

The default protocol is contemporaneous spatial nowcasting. At each block, the 800 visible targets are assimilated before predictions are made at the 200 held-out locations. Historical raw observations are not revisited. Whole-field future forecasting, in which no current target is visible, is outside the scope of the present evidence and is not included in the main comparison.

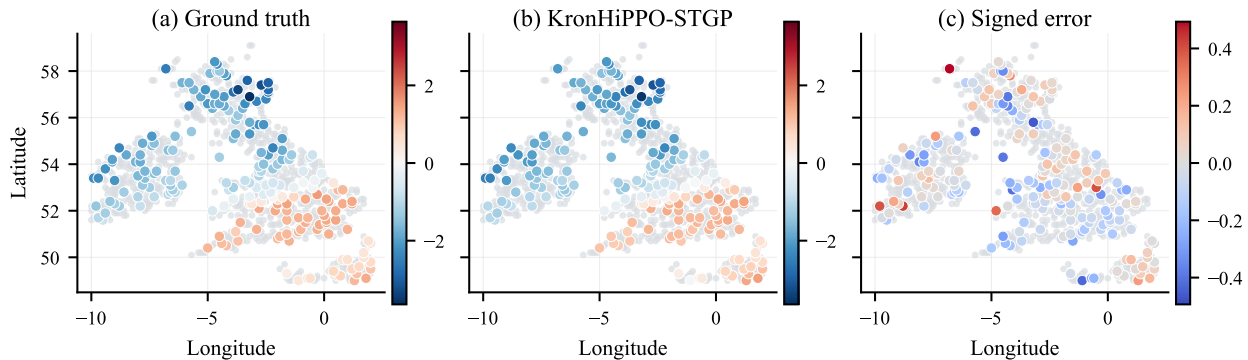


Figure 4: ERA5-Land Tasks 1–10 long-stream spatial snapshot at the final online block for the predeclared seed-0 split. The panels show the standardized target field, the KronHiPPO-STGP posterior mean, and the signed prediction error at the 200 held-out UK locations; the remaining 800 visible locations are shown in light gray for spatial context. The map geometry is used only for visualisation.

B.4 COVID-19

The COVID-19 dataset contains 52 weekly jurisdiction-level hospital-admission series from the CDC NHSN Hospital Respiratory Data release (Centers for Disease Control and Prevention, 2024). The target is the standardised $\log(1 + x)$ transform of weekly confirmed COVID-19 admissions per 100,000 population. Task 1 provides the initial 52-week calibration period, followed by 143 weeks processed strictly online. Each formal split observes 42 jurisdictions and predicts 10 held-out jurisdictions before their labels are revealed. The composite figure uses the predeclared seed-5 Connecticut and Nevada trajectories, selected by the largest distance between z-scored observed trajectories among the four archived candidates.

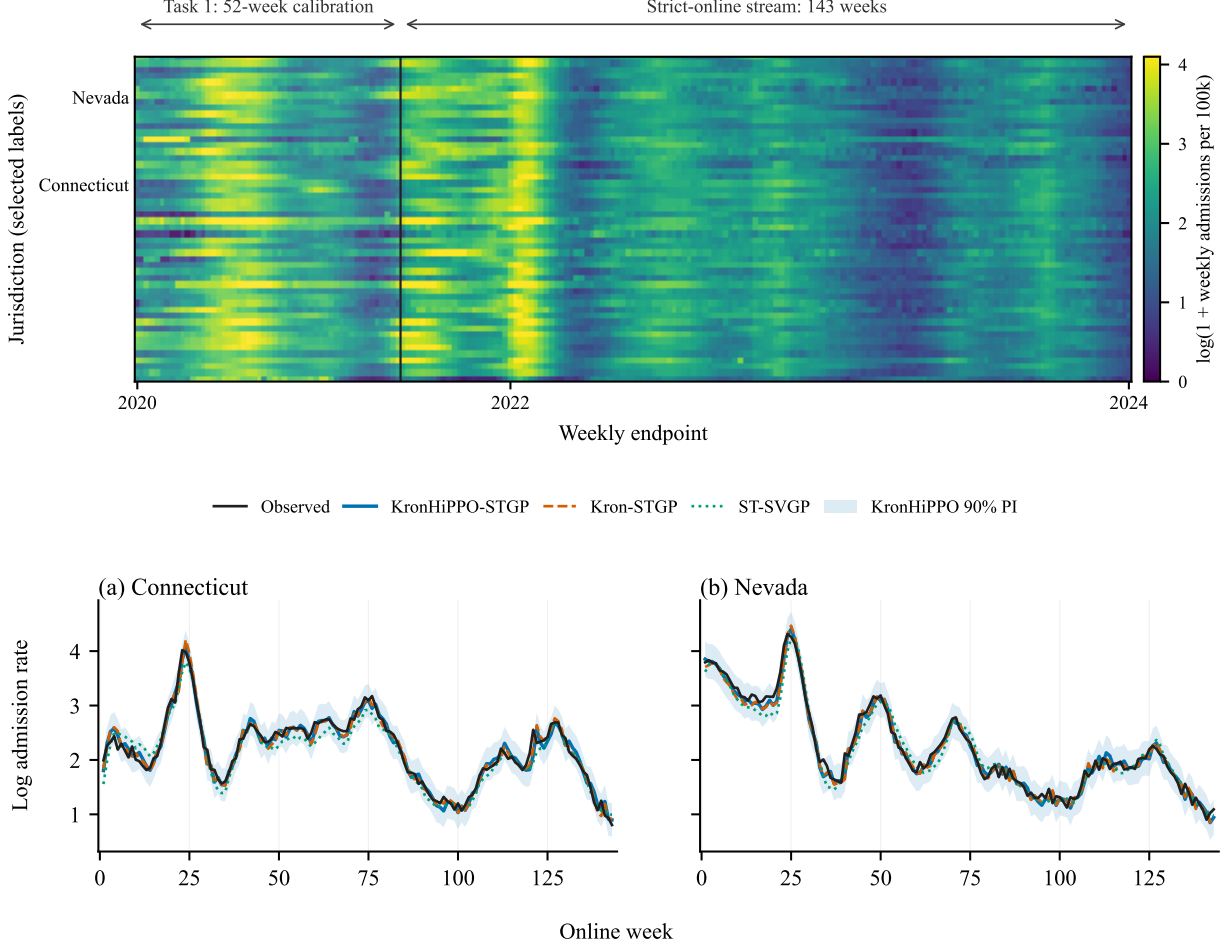


Figure 5: COVID-19 dataset structure and replay-free qualitative evidence. The upper panel shows the jurisdiction-by-week overview, with the 52-week calibration period separated from the subsequent 143-week strict-online stream. The lower panels show seed-5 Connecticut and Nevada trajectories selected by the archived observed-trajectory diversity rule; a shared y-scale and legend are used for the same predictive methods in both jurisdictions.

B.5 PEMS-BAY

PEMS-BAY contains traffic-speed measurements from 325 Bay Area sensors sampled every five minutes in the public DCRNN benchmark release (Li et al., 2018). We use the data for contemporaneous spatial nowcasting rather than future traffic forecasting. In each formal split, 260 sensors are visible and 65 are held out, with held-out labels revealed only after prediction. The protocol uses $M_s = 32$, $M_t = 128$, 512 random Fourier features, and the locked road-context X-lag mean design. All nine matched mechanism runs complete without OOM or divergence. Figure 2 uses the predeclared seed-1 sensors nearest the 15th, 50th, and 85th percentiles of full-stream KronHiPPO-STGP RMSE: sensors 400952, 400965, and 400178, respectively. Sensor selection therefore follows the predefined KronHiPPO-STGP difficulty quantiles. The common 48-hour plotting window is selected independently using observed speeds only, by maximising the median held-out speed range; the relative performance of the displayed methods is not used for window selection.

B.6 Mechanism and transfer diagnostics

Within each dataset, the joint/changing and zero-cross/changing variants share the same split, calibration, inducing-variable budgets, and changing inducing-representation maps. Zero-cross removes the accumulated trend-residual likelihood cross block after the transported historical and current-block statistics are combined.

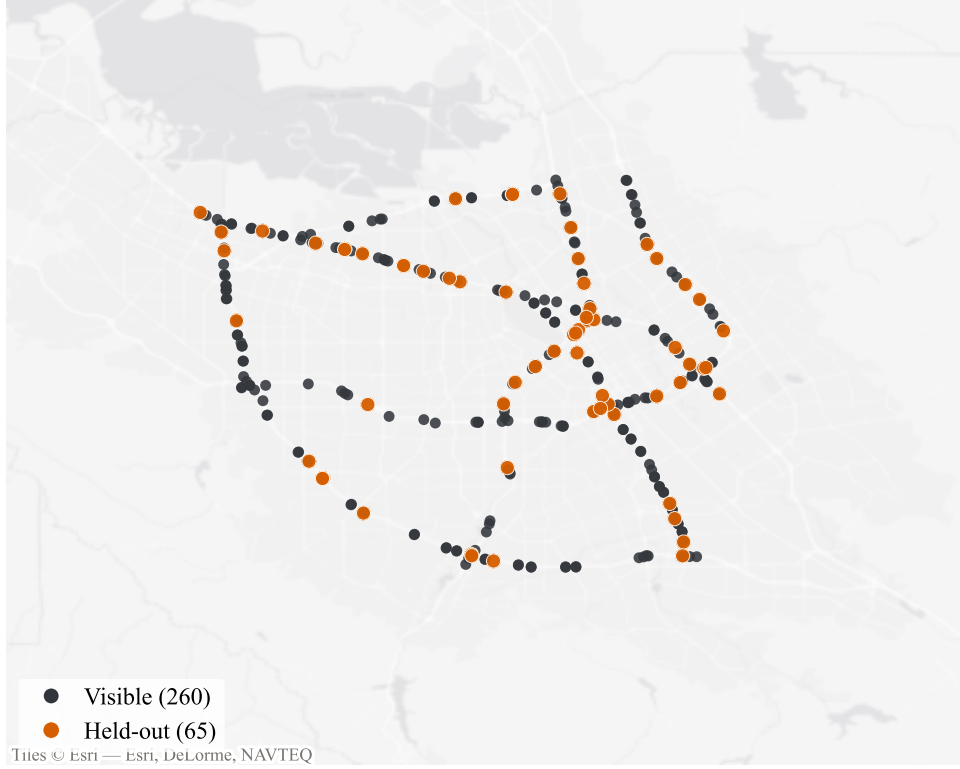


Figure 6: PEMS-BAY sensor geometry for the archived seed-0 spatial split. Dark-gray points are visible sensors and orange points are held-out sensors, shown over a pale Bay Area road basemap. The geometry is used only to visualise the contemporaneous held-out-sensor protocol. Map attribution is included in the artwork.

Joint/fixed-global uses complete-horizon temporal inducing representations and is reported separately as an offline representation reference.

The transfer diagnostic evaluates a separate mechanism by comparing the transported online state with a finite-dimensional variational reference posterior recomputed from the observed history in the current inducing representation. We report the marginal predictive Gaussian KL $KL(p_{\text{online}} \| p_{\text{reference}})$, averaged first over query locations within each non-warm-up block, then over blocks within each seed, and finally across seeds. At $M_t = 128$, the reported value 1.27×10^{-4} is this aggregated diagonal predictive KL; it is neither a joint-output KL nor a solver residual. The solver benchmark instead uses identical positive-definite systems and right-hand sides for dense and structured posterior solves, thereby isolating numerical solution from representation transfer.

For each solver size, dense and structured posterior solves are evaluated on the same matched random systems. The displayed dense and structured runtimes are means, whereas speed-up is computed within each replicate as $t_{\text{dense}}/t_{\text{structured}}$ before averaging across replicates. The resulting mean speed-up therefore need not equal the ratio of the two displayed mean runtimes. Figure 3 (c) visualises this matched solver benchmark, while Table 8 reports selected runtimes, paired speed-ups, and relative solution errors.

The lower off-diagonal block is reconstructed as the transpose when assembling the symmetric precision; all other retained statistics and assimilation steps remain unchanged. Because the removed block contains both historical and current-block contributions, the ablation tests joint likelihood coupling rather than historical trend revision alone.

Operating-point robustness. We repeat the zero-cross intervention under the archived ERA5 setting used in Table 1, without retuning either arm, to test whether the result depends on the spectral-mixture mechanism configuration. The full run reproduces the archived predictive means and variances to numerical precision (maximum absolute differences 4.23×10^{-12} and 5.15×10^{-15}), while zero-cross again degrades every point and

proper scoring-rule metric. Complete results are reported in Table 6.

Table 6: Full mechanism results and ERA5 operating-point robustness check (mean $\pm$ sample SD). The first two blocks report the formal ERA5-Land and PEMS-BAY mechanism suites; the final block repeats the joint-versus-zero-cross intervention under the Table-1 ERA5 configuration. Coverage90 provides an additional calibration diagnostic.

Variant	RMSE $\downarrow$	CRPS $\downarrow$	NLPD $\downarrow$	ECE $\downarrow$	Coverage90
<i>ERA5-Land (five 171-block runs)</i>					
Joint / changing	0.1251 $\pm$ 0.0034	0.0665 $\pm$ 0.0014	-0.6837 $\pm$ 0.0186	0.0461 $\pm$ 0.0029	0.9134 $\pm$ 0.0035
Zero-cross / changing	1.2898 $\pm$ 0.0120	0.9951 $\pm$ 0.0088	52.8139 $\pm$ 1.9682	0.6905 $\pm$ 0.0018	0.1148 $\pm$ 0.0019
Joint / fixed-global	0.1950 $\pm$ 0.0030	0.1420 $\pm$ 0.0009	0.2580 $\pm$ 0.0043	0.1807 $\pm$ 0.0012	0.9933 $\pm$ 0.0006
<i>PEMS-BAY (three formal spatial splits)</i>					
Joint / changing	0.7666 $\pm$ 0.0629	0.3774 $\pm$ 0.0193	1.1883 $\pm$ 0.1076	0.1317 $\pm$ 0.0196	0.9032 $\pm$ 0.0204
Zero-cross / changing	0.9555 $\pm$ 0.0232	0.4856 $\pm$ 0.0123	1.5333 $\pm$ 0.1323	0.0474 $\pm$ 0.0132	0.8490 $\pm$ 0.0217
Joint / fixed-global	0.8112 $\pm$ 0.0665	0.4092 $\pm$ 0.0208	1.2194 $\pm$ 0.0637	0.1760 $\pm$ 0.0101	0.9249 $\pm$ 0.0112
<i>ERA5-Land (Table-1 operating point)</i>					
Joint / changing	0.1416 $\pm$ 0.0036	0.0766 $\pm$ 0.0017	-0.3349 $\pm$ 0.0434	0.0669 $\pm$ 0.0067	0.7830 $\pm$ 0.0089
Zero-cross / changing	1.2921 $\pm$ 0.0127	1.0142 $\pm$ 0.0090	97.1323 $\pm$ 2.8269	0.4599 $\pm$ 0.0007	0.0841 $\pm$ 0.0013

Table 7: Coordinate-transfer control at $M_t = 128$ (mean $\pm$ sample SD over five spatial splits). Predictive KL is $\text{KL}(p_{\text{online}} \| p_{\text{reference}})$ against the finite-dimensional variational reference posterior recomputed in the current inducing representation. Identity reuse changes only the historical likelihood transfer; all current-representation model and inference components are held fixed. Coverage90 has a nominal target of 0.90.

Metric	Conditional transport	Identity reuse
Predictive KL $\downarrow$	$1.2649 \times 10^{-4} \pm 5.9122 \times 10^{-6}$	2.0713 ± 0.1158
RMSE $\downarrow$	0.0795 ± 0.0051	0.2651 ± 0.0024
CRPS $\downarrow$	0.0463 ± 0.0017	0.1520 ± 0.0015
NLPD $\downarrow$	-0.9687 ± 0.0251	1.0741 ± 0.0960
ECE $\downarrow$	0.1913 ± 0.0085	0.1604 ± 0.0055
Coverage90	0.9825 ± 0.0038	0.6320 ± 0.0074

Table 8: Dense versus Schur–Sylvester posterior solves on matched systems. Runtimes are reported as means; speed-up is the mean of the within-replicate $t_{\text{dense}}/t_{\text{structured}}$ ratios and therefore need not equal the ratio of the mean runtimes. Relative error is measured against the dense solution.

$M_s = M_t$	Dense (ms)	Structured (ms)	Speed-up $\uparrow$	Rel. error $\downarrow$
16	1.072	0.137	7.9 $\times$	9.2×10^{-14}
32	27.919	0.403	69.6 $\times$	9.1×10^{-13}
64	994.822	0.974	1036.7 $\times$	4.7×10^{-12}

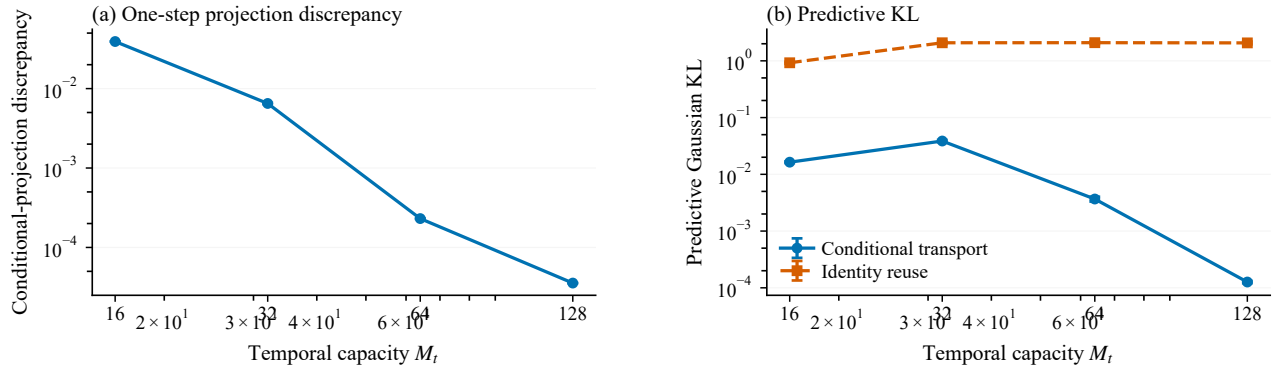


Figure 7: Transfer diagnostics. (a) One-step conditional-projection discrepancy. (b) Predictive Gaussian KL to the current-representation reference across temporal capacities for conditional transport and identity reuse.

Conditional transport is non-monotone across the sweep, while identity reuse remains at order-one KL for $M_t \geq 32$.

B.7 Implementation efficiency audit

The archived short-batch implementation audit at $M_s = M_t = 128$ on an RTX 4090 reports the ST-SVGP and KronHiPPO-STGP accuracy, training wall time, profiler count, and peak reserved GPU memory from the same audit record. This is an implementation-level comparison rather than a framework-neutral complexity benchmark: the profiler column reports supported-operation counts from the corresponding implementations and does not represent complete hardware instruction-level FLOPs. The reported $40.4\times$ difference is therefore interpreted as a profiler-count ratio, while the wall-clock and peak-memory ratios are measured implementation outcomes.

Table 9: Archived ERA5 short-batch implementation audit at $M_s = M_t = 128$ on an RTX 4090. Accuracy and efficiency entries within each row come from the same audit record. **Train** denotes recorded training wall time, **Profiler GFLOPs/step** denotes supported-operation profiler counts, and **Peak** denotes reserved GPU memory.

Method	RMSE	Train (s)	Profiler GFLOPs/step	Peak (GiB)
ST-SVGP	0.0837	4,586.83	1,106.04	14.75
KronHiPPO-STGP	0.0882	41.39	27.36	1.68
<i>Relative difference</i>	+5.4% RMSE	110.8 $\times$ faster	40.4 $\times$ lower	8.8 $\times$ lower

Appendix Table 10 reports a separate strict-online accounting over Tasks 2–10, including end-to-end execution, per-block posterior-update latency, prediction time, GPU memory, and persistent-state size. These online records use a different measurement scope from the short-batch audit and are therefore not combined with Table 9 to form a cross-hardware or architecture-independent speed ratio.

Table 10: Online efficiency summary for ERA5 Tasks 2–10 using the formal method names. End-to-end timing includes the online blocks, while the GPU and persistent-state columns are implementation-specific. These values should not be interpreted as hardware-normalised cross-method speed ratios.

Method	End-to-end (s)	Iter./block (s)	Predict (s)	GPU MiB	State MiB
OSGPR	78.00	0.1923	20.270	657	0.25
StreamingSGPR	29.81	0.0336	1.414	816	0.25
Official OHSVGP	254.32	1.2312	12.398	1966	0.08
KronHiPPO-STGP	79.52	0.0214	3.562	1067	19.16
Kron-STGP	61.06	0.0204	3.500	1069	20.92